

Upanshi Mittal

Full Stack Developer , ML engineer

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PROFESSIONAL SUMMARY

AI-focused Full Stack Developer with hands-on experience building intelligent, scalable web applications. Proficient in the MERN stack, Python, and integrating machine learning models into production-ready systems. Demonstrated ability to deliver end-to-end solutions across accessibility tools, AI-driven platforms, and real-time data systems. Eager to contribute to high-impact engineering teams through internship or full-time roles.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, C++, SQL

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, GSAP, Framer Motion

Backend: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication

Databases: MongoDB, MySQL, SQLite

AI / ML: Scikit-Learn, OpenCV, NLP, Generative AI, Vertex AI, Sentence Transformers, MediaPipe

DevOps & Tools: Git, GitHub, Docker, Firebase, Postman, pytest

PROJECTS

- AccessLens — AI-Based Accessibility Enhancement Tool** | *Python · FastAPI · Next.js · LLaVA · Mistral 7B · Redis · Docker*
 - Built an **AI-powered accessibility platform** to assist visually impaired users with real-time object and text detection, improving environmental awareness by leveraging LLaVA and Mistral 7B multimodal models.
 - Engineered a **high-performance backend** with FastAPI and Redis caching, reducing API response latency and supporting concurrent user sessions via Docker containerization.
 - Designed an **intuitive Next.js** + Tailwind CSS frontend with Framer Motion animations, achieving seamless UX and meeting WCAG digital accessibility standards.
 - Validated system reliability with a comprehensive **pytest test suite**; integrated Playwright for end-to-end browser automation testing.
- Intervue — AI-Powered Interview Preparation & Evaluation System** | *Next.js · Vertex AI · ElevenLabs · TailwindCSS · Pydantic · Python*
 - Developed an AI-driven interview simulation platform using **Vertex AI for NLP evaluation**, enabling real-time scoring of candidate responses across HR and technical rounds.
 - Integrated **ElevenLabs voice synthesis** to create realistic, voice-interactive interview sessions, enhancing candidate preparation realism and engagement.

- Designed **structured evaluation pipelines** with Pydantic data validation, delivering actionable performance insights and reducing manual hiring review time.
3. **Callarity — Smart Communication & Call Analysis System** | *Next.js · Node.js · Python · Deepgram · Ollama · MongoDB*
- Built an end-to-end **call analytics system** using Deepgram speech-to-text APIs and Ollama LLMs for real-time sentiment detection and performance insights.
 - Designed **scalable MongoDB backend** services to store, query, and retrieve call transcripts and analytics data efficiently at scale.
 - Delivered **actionable communication clarity metrics** through a clean Next.js dashboard, improving feedback loops for sales and support teams.
4. **BeatFall — Rhythm-Based 3D Game** | *React · React Three Fiber · MediaPipe · OpenCV · Firebase*
- Built an **immersive 3D rhythm game** using React Three Fiber with optimized rendering pipeline to achieve low-latency, real-time gameplay performance.
 - Integrated **MediaPipe and OpenCV** for gesture-controlled mechanics, enabling hands-free interaction that enhanced accessibility and gameplay novelty.
 - Implemented **Firebase real-time database** for live score tracking and multi-user session management, supporting concurrent players without data race conditions.

EXPERIENCE

Open Source Contributor · Open World Holidays Framework

Contributed feature additions and bug fixes to the open-source Open World Holidays Framework, practicing collaborative development workflows via Git pull requests and code reviews.

Tech Head · Robotics Hub, IIIT

Led technical initiatives and mentored club members on robotics and software development projects, fostering hands-on learning culture within the student community.

Frontend Intern · InAmigos NGO | 2-week Internship

Built responsive frontend components using React and Tailwind CSS.

ACHIEVEMENTS & RECOGNITION

1st Place — All-Girls Team, College Inter-Department Coding Competition (competitive programming).

Best Negotiator Award — Startup Showdown, IEEE IIIT Chapter.

Special Mention — FOSS Hack (national open-source hackathon).

EDUCATION

B.Tech — Computer Science & Engineering · Jaypee Institute of Information Technology (IIIT), Noida

Class XII (PCM + CS) · Jaypee Vidya Mandir

Class X · Jaypee Vidya Mandir

LANGUAGES

English (Professional Proficiency) · Hindi (Native)